

Solution Requirements

Date	26 June 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Registered users can create an account, sign in securely, and access the application using verified credentials. Session management ensures that only authenticated users can use the platform.	High
2	Authorization levels	Different access privileges are assigned based on user roles. Administrators can manage users, monitor predictions, and access analytics, while regular users can perform flood predictions, view prediction history, and update their profiles.	High
3	External interfaces	The application communicates with the trained machine learning model and SQLite database through the Flask backend. It is also designed to support integration with weather APIs and other external data sources in future versions.	Medium

4	Transactions processing	Registered users can create an account, sign in securely, and access the application using verified credentials. Session management ensures that only authenticated users can use the platform.	High
5	Reporting	User inputs are validated and preprocessed before being passed to the prediction model. After prediction, the generated result is stored in the database and displayed instantly on the dashboard.	Medium
6	Business rules, etc.	Flood predictions are available only after successful login. All mandatory input fields must be completed before prediction, and every prediction is recorded with the corresponding user and timestamp for future reference.	High
7	Compliance to laws or regulations	User information is handled securely by following standard security practices such as password protection, controlled access, and safe database management to maintain data privacy.	Medium
8	Other	The project architecture supports future improvements such as cloud deployment, live weather data integration, GIS-based flood mapping, SMS/email notifications, and periodic model retraining.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	The system must execute model preprocessing and XGBoost prediction inferences swiftly to provide instant feedback to meteorologists during emergencies.	Average prediction API inference response times must remain under 100 milliseconds (78ms achieved in load testing).
2	Scalability	Handle spikes in traffic during storm surges without throwing database lock exceptions or dropping requests.	Sustain a concurrent load of 50 virtual users executing predictions with over 70 requests/sec throughput and 0% error rates.
3	Security & Data Privacy	Secure sensitive database entries, user credentials, and session access points against common security vulnerabilities.	Hash password credentials using bcrypt with a salt work factor of 12. Enforce secure, encrypted HTTP session cookies.
4	Reliability & Availability	Maintain consistent web service availability during severe weather events.	Ensure a target web hosting uptime of 99.9% on Render, with automated database connection recovery via WAL pragmas.
5	Usability & Accessibility	Ensure the frontend interface is intuitive, fully responsive, and loads fast on mobile and desktop viewports.	Page load times under 2 seconds. Responsive grid layout conforming to WCAG 2.1 AA standards for colors and readability.

6	Other	Minimize model binary disk size and runtime memory footprints to prevent cloud hosting cold start limits.	Serialized model file size under 500KB (483KB). Memory usage under 200MB during concurrent inference serving.
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